Korea Technology and Information Promotion Agency for SMEs

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Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Ammonia-Ready Burner System with low NOx			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen (√), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Mechanical and Materials		Energy & Resources	
	Medium Category	Small Category	Medium Category	Small Category
	Energy/ Environment Mechanical Systems	Boilers and Furnace Equipment	New & Renewable Energy	Hydrogen
Concept of Proposed Technology	Ammonia is an attractive energy carrier because of its high hydrogen content and its potential to replace conventional fuels as a carbon-free alternative. Its production, storage and transport are more mature and convenient than those for hydrogen. However, ammonia combustion presents significant challenges: it is difficult to ignite, has low flame velocity and a narrow flammability range, and tends to generate relatively high nitrogen-oxide (NO_x) emissions. So based on a thin-flame combustor concept archiving low NO_x emission while having a stable combustion is a promising solution for Carbon free combustion. This process is needed in many different industry fields.			

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Rationale for Joint R&D Project	Fraunhofer UMSICHT possesses comprehensive test infrastructure capable of operating both pure fuels and arbitrary mixtures of natural gas, hydrogen, and ammonia. Moreover, this work would serve as an ideal continuation of the W.BM-H2 Subproject 3 research, which focuses on hydrogen. Developing a thin flame burner system that can run on CO_2-free fuels or on blends that include CO_2-free components is critical for achieving meaningful decarbonization in combustion applications. As Fraunhofer UMSICHT does not manufacture burners, collaboration with a suitable burner manufacturer is essential to translate research outcomes into deployable hardware. Fraunhofer UMSICHT will provide test rigs equipped with exhaust-gas analysis instruments, a flexible fuel supply system, and a combustion chamber, all suitable for burners up to 100 kW. This partnership enables rapid experimental validation, optimization of low-NO_x combustion strategies for ammonia and hydrogen blends, and accelerated scaling toward industrial implementation.
Key Words	1. Ammonia-ready thin-flame combustor compact burner engineered for stable combustion on ammonia and ammonia blends. 2. Low NO_x operation, thin flame, lean/distributed combustion reduces peak temperatures and avoids local oxygen rich hotspots, thereby suppressing thermal (Zeldovich) NO_x formation, which is highly temperature sensitive. 3. Off grid process heat capability, provides process heat independent of electrical infrastructure, making the system suitable for remote, maritime or standalone industrial sites. 4. Market and demand assessment, evaluate demand and deployment scenarios for carbon free ammonia combustion, leveraging ammonia's more mature production, storage and transport infrastructure compared with hydrogen. 5. carbon-free combustion and with more mature storage and transportation than those for hydrogen

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (TRL3-4)	End of the Project	Level (TRL6)
R&D Objectives	The primary objective of this project is to study the market and technical need for a carbon-free combustor and to develop a thin-flame burner system that operates on ammonia while meeting stringent ultra-low NO_x limits. Building on the experience of Korean KMUs that already produce ultra-low NO_x combustors for natural gas, the project will adapt and extend those combustion concepts to address the specific challenges of ammonia, difficult ignition, low flame speed, narrow flammability limits and a tendency to form NO_x. Specifically, the R&D work will: • evaluate use cases and market demand for ammonia-fired burners and define performance and emissions targets • develop an optimized thin-flame burner core and fuel-flexible injection system that ensure stable ignition and flame propagation across ammonia blends and pure ammonia • create advanced control strategies and emissions-mitigation measures to achieve certified low-NO_x performance			
R&D Target Specifications	Successful completion of these objectives will deliver a validated prototype combustor ready for industrialization and certification, demonstrating that ammonia can be used as a practical, low-carbon fuel in applications that require both high reliability and compliance with strict NO_x limits. here are Korean KMUs which have ultra-low NOx combustors for the use with natural gas. To study the need of combustor whose combustion can be carbon free and develop such a combustor which is full filling Nox limitations Is a The Objective to develop an.			
R&D Plan	The R&D project will begin with Phase I (Pre-study, Months 0–6) during which a mutual NDA and consortium agreement will be drafted and signed by all partners to secure confidentiality, IP rules and collaboration terms, and a market pre-study will be conducted to assess future ammonia availability and the future demand for ammonia burner systems. Evaluation of potential use cases and deployment scenarios for example industrial process heat, district heating, retrofit opportunities for existing gas burners, and maritime or remote-site applications. In Year 1 the burner manufacturer will complete the detailed design and build the thin-flame combustor system, deliver it to Fraunhofer UMSICHT, and support installation while finalizing interfaces, safety requirements and any necessary test-rig adaptations. Year 2 will focus on commissioning and the first experimental campaigns: after safety clearance, progressive tests with fuel mixtures containing up to 20 vol% ammonia will be carried out, with emissions measurements, efficiency analyses and control-system tuning feeding back into iterative design improvements. Year 3 will concentrate on further testing and optimization to enable operation on 100 % ammonia, including long-duration endurance runs, reliability assessments and preparation of the technical documentation required for certification and industrialization. • Months 0 - 6 Phase I: **Pre-study, NDA & Consortium Agreement**: Draft, negotiate and sign a mutual NDA and consortium agreement; perform market research on future ammonia availability and demand; evaluate deployment use cases and application requirements for ammonia burner systems; and define project scope, safety framework and initial test requirements. • Year 1 **Design, Manufacture & Delivery**: Complete detailed design, fabricate the thin-flame burner system, perform factory acceptance checks and deliver/install the system at Fraunhofer UMSICHT; finalize interfaces and test-rig			

	adaptations. • Year 2 **Commissioning & Test Campaigns** (up to 20 vol% NH_3): Commission the test rig, conduct progressive experimental campaigns with fuel blends up to 20 vol% ammonia, perform emissions and performance analyses, and implement interim design and control optimizations. • Year 3 **Extended Testing & 100% NH_3 Demonstration**: Execute extended endurance and reliability tests, progressively increase ammonia share toward 100% operation, finalize optimization, and compile the technical dossier for certification and commercialization.

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	UMSICHT will act as the project's test and validation partner and will lead all experimental activities required to develop and qualify the thin-flame ammonia combustor. Key responsibilities include: • Test-rig adaptation and integration: Implement any mechanical, electrical or instrumentation modifications to the existing test rig to fully accommodate the proposed combustor, including mounting interfaces, fuel supply connections and exhaust handling. • Fuel procurement and handling: Procure and manage all fuels required for the test campaigns (natural gas, hydrogen and ammonia blends), establish safe storage and handling procedures, and coordinate logistics for fuel delivery and on-site supply. • Commissioning and instrumentation: Commission the installed burner, install and calibrate sensors and measurement systems (flow, pressure, temperature, emissions), and ensure data-acquisition systems are configured for needed measurements. • Test execution and troubleshooting: Conduct the experimental campaigns from commissioning through endurance testing, provide on-site test operation, monitor combustion stability and emissions, and work with the Korean partner to diagnose and resolve technical issues during tests. • Data analysis and performance evaluation: Process and analyze test data (efficiency, NO_x and NH_3-slip), generate quantitative assessments, and identify design and control improvements. • Iterative optimization support: Feed experimental findings back to the burner manufacturer to guide iterative design changes, control-system tuning and subsequent test matrices. • Reporting and documentation: Produce test reports, data records and a consolidated final report; recommended design modifications and validated operating envelopes protocols.
Expected R&R of Korean SME Partner	The Korean KMU will serve as the industrial burner partner and support the joint R&D activities across all project phases. Including the following: • Design and engineering: Lead detailed engineering of the thin-flame combustor, deliver CAD models and, where required, CFD analyses, define mechanical, thermal and control interfaces, and provide the drawings and specifications necessary for manufacture and test-rig integration. • Manufacturing and prototyping: Develop and manufacture the burner and its control system, produce prototype units, and iteratively refine the design based on experimental findings. • Test support and troubleshooting: Provide on-site and remote technical support during installation, commissioning and experimental campaigns; diagnose issues encountered during testing and implement design modifications informed by test results. • Data sharing and reporting: Prepare and deliver engineering reports, test plans and interim technical reports; participate in milestone reviews and document design changes and lessons learned in a timely manner. • Market and commercialization input: Supply market intelligence from Korean and international customers, advise on manufacturability and planning for commercialization. • Consortium participation and IP cooperation: Join the NDA and consortium agreement, actively engage in project meetings, and collaborate on IP management, publication approvals and exploitation plans in accordance with consortium rules.

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	The project will deliver an approximately 100 kW thin-flame burner system specifically optimized for ammonia and ammonia-blend fuels. Through iterative design, comprehensive testing at Fraunhofer UMSICHT's facilities, and close collaboration with the Korean SME partner, the project will produce a validated prototype ready for industrialization and certification. The resulting technology is expected to evolve into a commercially marketable ammonia burner, offering a low-NO_x combustion solution that advances decarbonization goals and opens new market opportunities for clean-fuel applications.

6. References

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization

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